

WASP-43b

R-0645

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_170324 · created 2026-07-25T20:17:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457836.6925 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.146 +/- 0.055
Transit depth (Rp/Rs)^2	2.14 +/- 1.6 [%]
Orbital Inclination (inc)	87.1 +/- 2.94
Airmass coefficient 1 (a1)	0.942 +/- 0.076
Airmass coefficient 2 (a2)	0.054 +/- 0.051
Scatter in the residuals of the lightcurve fit is	8.21 %
Best Comparison Star	#3 - [220, 152]
Optimal Method	PSF photometry
Transit Duration (day)	0.0585 +/- 0.0053

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.146 ± 0.055 vs 0.1594 (deviation 0.2σ)
Duration fit vs published	0.0585 d vs 0.0512 d (deviation 1.11σ)
Mid-time O-C	-3.1 min
Depth SNR	2.2
Residual scatter	8.21 %

Light curve

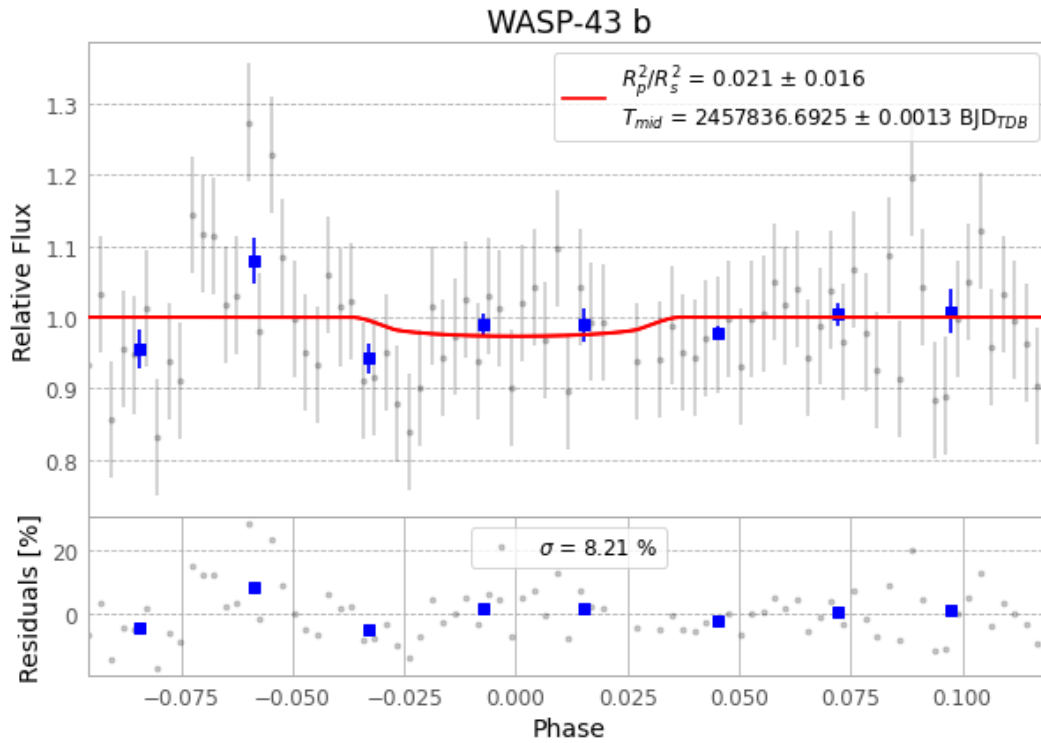


Figure 1 — FinalLightCurve_WASP-43 b_2017-03-23.png (SHA-256 dd9dad9f71bbf3e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [241, 163], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c141e1277a74dcc5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

85 FITS frames (56 MB), WASP-43170324023612.FITS → WASP-43170324064812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-170324.log (a323d8607787...), FinalLightCurve_WASP-43 b_2017-03-23.png (dd9dad9f71b...), FinalLightCurve_WASP-43 b_2017-03-23.csv (e2d464e81a04...)

Compute resources

Wall time 185.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 20:17Z.